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Course - B. Tech (Electronics & Electrical Engineering)

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Abstract

Static Random Access Memory (SRAM) is a high-speed volatile memory widely used in modern digital systems for applications such as cache memory and embedded processors. This report presents the design and simulation of a 6-transistor (6T) SRAM memory cell using CMOS technology. The proposed design utilizes two cross-coupled inverters to store a single bit of data and two access transistors controlled by a word line to perform read and write operations through complementary bit lines.

The working of the SRAM cell is analyzed under three modes of operation: hold, read, and write. Proper transistor sizing and circuit configuration are considered to ensure data stability and reliable operation. The simulation of the SRAM cell is carried out using electronic design automation tools, and the results verify correct functionality, showing stable data retention and accurate switching behaviour during read and write operations.

Furthermore, the report discusses important design considerations such as static noise margin, power consumption, and area optimization. The advantages, limitations, and practical applications of SRAM are also highlighted. The study concludes that SRAM provides fast access time, high reliability, and efficient performance, making it a critical component in high-speed memory systems.

Introduction

Memory plays a vital role in modern digital systems, as it is used to store data and instructions required for processing. Among various types of semiconductor memories, Static Random Access Memory (SRAM) is widely used due to its high speed, reliability, and simple operation. Unlike Dynamic Random Access Memory (DRAM), SRAM does not require periodic refreshing, which makes it faster and more efficient for applications where quick data access is essential.

SRAM stores data using bistable circuits formed by cross-coupled inverters, allowing it to retain information as long as power is supplied. The most commonly used SRAM configuration is the 6-transistor (6T) SRAM cell, which consists of two CMOS inverters and two access transistors. These access transistors are controlled by a word line and are connected to complementary bit lines, enabling read and write operations.

Due to its fast access time and stability, SRAM is primarily used in cache memory of microprocessors, registers, and embedded systems. However, it occupies more area and is more expensive compared to DRAM, which limits its use in large memory applications.

This report focuses on the design and simulation of a 6T SRAM memory cell using CMOS technology. It explains the circuit structure, working principles, and simulation results, along with key design considerations such as stability, power consumption, and noise margin. The objective is to understand the behaviour of SRAM cells and evaluate their performance in practical applications.

➤ Basics Concept of RAM

Random Access Memory (RAM) is a type of semiconductor memory used to store data and instructions temporarily while a computer or digital system is operating. It is called random access because any memory location can be accessed directly in equal time, without following a sequential order.

RAM is a volatile memory, which means it loses all stored data when the power supply is turned off. It acts as a working memory for the system, allowing fast read and write operations that are essential for efficient processing.

The 6T SRAM cell consists of:

- Two cross-coupled CMOS inverters
- Two NMOS access transistors
- Bit lines (B and B')
- Word line (WL)

The cross-coupled inverters store the data, while access transistors allow read/write operations.

➤ Types of RAM

RAM is mainly classified into two types:

1. Static RAM (SRAM)

- Stores data using flip-flop circuits (bistable latches)
- Does not require refreshing
- Faster and more stable
- Used in cache memory

2. Dynamic RAM (DRAM)

- Stores data in capacitors
- Requires periodic refreshing
- Slower but more compact
- Used in main memory

➤ Simulation of SRAM Memory Cell

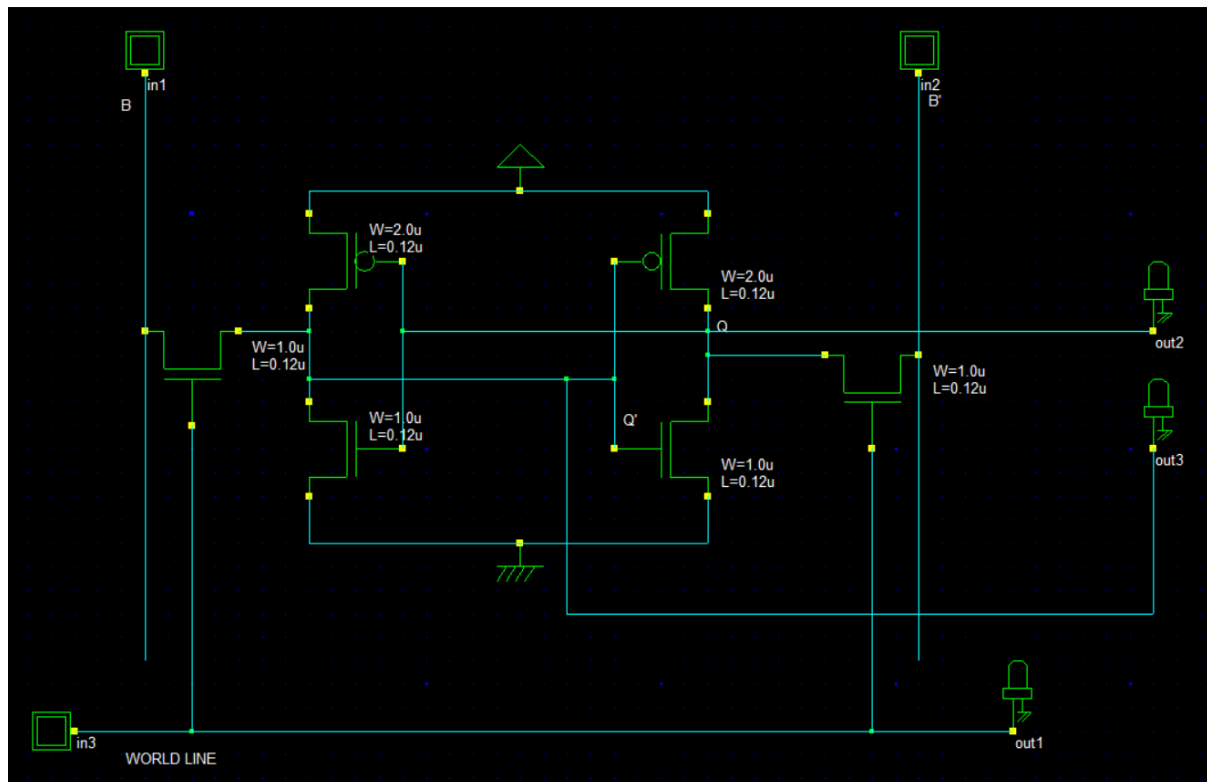
The simulation of the 6-transistor (6T) SRAM memory cell is performed to verify its functionality and analyze its performance under different operating conditions. The circuit is designed using CMOS technology and simulated using tool DSCH.

Simulation Setup

The SRAM cell consists of two cross-coupled inverters forming a bistable latch and two NMOS access transistors controlled by the word line (WL). The cell is connected to complementary bit lines (B and B'), which are used for data transfer. A DC power supply (VDD) and ground are applied to the circuit, and input signals are provided to the word line and bit lines. The outputs are observed at nodes Q and Q'.

Simulation Procedure

The simulation is carried out by applying input signals and observing the output waveforms. First, the SRAM cell is designed using CMOS transistors. Then, power supply and ground connections are applied. Input signals are given to WL, B, and B', and transient analysis is performed to study the time-based behaviour of the circuit.



➤ Simulation Analysis

The behaviour of the SRAM cell is analyzed under three operating conditions:

Hold Operation

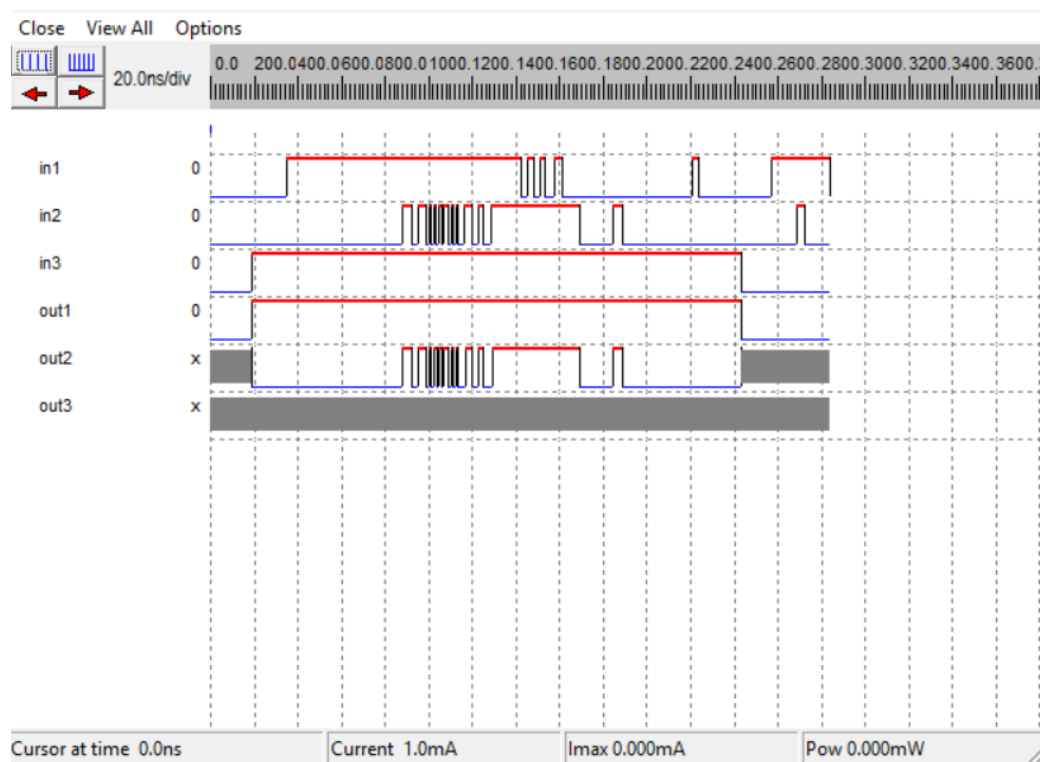
When the word line (WL) is low, the access transistors remain OFF, isolating the cell from the bit lines. The stored data is maintained by the cross-coupled inverters without any disturbance.

Write Operation

When the word line (WL) is high, the access transistors turn ON, allowing data from the bit lines (B and B') to be written into the cell. The previous data is overwritten, and the outputs Q and Q' change according to the input.

Read Operation

During the read operation, the word line is activated, and the stored data affects the bit lines. A small voltage difference is generated, which can be detected without disturbing the stored value.



Graph Analysis

The waveform illustrates the behaviour of the SRAM cell under different input conditions. The signals in1, in2, and in3 correspond to the bit lines (B, B') and the word line (WL), respectively, while out1, out2, and out3 represent the output responses of the circuit. When the word line is low, the SRAM cell remains in the hold state, and the outputs retain their previous values, indicating stable data storage. When the word line becomes high, the access transistors turn ON, allowing data to be written into or read from the cell. During the write operation, changes in the bit lines directly affect the outputs, demonstrating successful data storage. During the read operation, the outputs reflect the stored value without altering the internal state of the cell. The waveform confirms that the SRAM cell operates correctly in all modes, including hold, read, and write.

WL	B	B'	Operation	Q (Next State)	Q'
0	X	X	Hold	No Change	No Change
1	1	0	Write 1	1	0
1	0	1	Write 0	0	1
1	1	1	Read	No Change	No Change

Truth Table (Output Analysis)

Note:

- $X = \text{Don't care}$
- Q' is always the complement of Q

Results and Discussion

The simulation results show that the outputs Q and Q' remain complementary at all times. The SRAM cell maintains stable data during the hold condition, performs accurate data writing, and allows reliable data reading. No data loss or unintended switching is observed, indicating proper circuit operation.

Conclusion

In this report, the design and simulation of a 6-transistor (6T) SRAM memory cell have been successfully presented and analyzed. The SRAM cell, based on CMOS technology, utilizes cross-coupled inverters to store data and access transistors controlled by the word line to perform read and write operations through complementary bit lines. The working of the SRAM cell was studied under different modes, including hold, read, and write, demonstrating its ability to store and maintain data reliably.

The simulation results, along with the waveform analysis, confirm that the designed SRAM cell operates correctly and efficiently. The outputs remain stable during the hold condition, update accurately during write operations, and allow reliable data sensing during read operations without disturbing the stored information. The complementary nature of the outputs ensures proper functionality of the memory cell.

Furthermore, important design aspects such as transistor sizing, stability, and noise margin were considered to ensure reliable performance. Although SRAM offers high speed and does not require refreshing, it occupies more area compared to DRAM, making it more suitable for applications where speed is critical rather than storage capacity.

Overall, the study demonstrates that SRAM is an essential component in modern digital systems, particularly in cache memory and high-speed applications. The successful design and simulation validate its effectiveness, stability, and suitability for use in advanced electronic systems.